Dipoles and Its Impacts

Light starts from Dipoles. One can say dipole is one of the most important pieces in Nature.



$\vec{P} = N(-q)\vec{r}$ N: dipole density.

Dipoles help building electronic transition, lattice vibrations, origin of light, ...
To understand why dipoles are origin of light, look at Macroscopic Maxwell's equations (Gauss unit)

$$\vec{\nabla}\cdot\vec{E} = -4\pi\vec{\nabla}\cdot\vec{P} + 4\pi\rho_{macro} \qquad \nabla\times\vec{E} = -\tfrac{1}{c}\tfrac{\partial}{\partial t}\vec{B} \qquad \vec{H} = \vec{B}$$
$$\vec{\nabla}\cdot\vec{B} = 0 \qquad \nabla\times\vec{H} = \tfrac{1}{c}\tfrac{\partial}{\partial t}(\vec{E} + 4\pi\vec{P}) + \underbrace{\tfrac{4\pi}{c}J_{macro}}_{0} \qquad \vec{D} = \vec{E} + 4\pi\vec{P}$$

Consider simple waves: $\vec{E}(\vec{r},t) = \vec{E}_T(\vec{r},t) + \vec{E}_L(\vec{r},t)$

T:  $\vec{\kappa}\cdot\vec{E}_T = 0$ L: → → → ← ← ← $\vec{\kappa}\times\vec{E}_L = 0$

In Isotropic medium: we have $\vec{E} = \vec{E}_T$, the maxwell's equations give

$$\left\{\tfrac{1}{c^2}\tfrac{\partial^2}{\partial t^2} - \vec{\nabla}^2\right\}\vec{E}(\vec{r},t) = -\tfrac{4\pi}{c^2}\tfrac{\partial^2}{\partial t^2}\vec{P}(\vec{r},t)$$ wave equation for transverse wave.

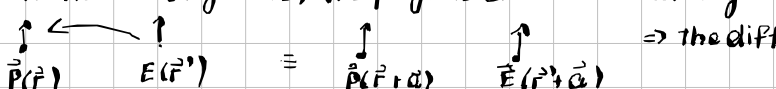
In optical response

$$\vec{P}(\vec{r},t) = \int_{all} d^3\vec{r}' \int_{-\infty}^{t} dt' \; \vec{\chi}(\vec{r},\vec{r}',t,t')\;\vec{E}(\vec{r}',t')$$ χ: susceptibility tensor

The response to the fields everywhere and all time contribute to the polarization
If the material is homogeneous in space and time, that means response is translation invariant

$$\chi(\vec{r}+\vec{a},\vec{r}'+\vec{a},t,t') = \chi(\vec{r},\vec{r}',t,t') \quad (\vec{r}+\vec{a})-(\vec{r}'+\vec{a}) = \vec{r}-\vec{r}'.$$

Since materials look identical everywhere, the physics should not change



⇒ the difference $\vec{r}-\vec{r}'$ matters

It is also time invariant, since materials at time t are not different to time $t+\Delta t$ Thus,

$$\chi(\vec{r},\vec{r}',t,t') = \chi(\vec{r}-\vec{r}',t-t')$$

Now, the response simply asks
how far away did the cause occur?
how long ago did the cause occur

$$\vec{P}(\vec{r},t) = \int d^3\vec{r}' \int_{-\infty}^{t} dt' \; \vec{\chi}(\vec{r}-\vec{r}',t-t')\,\vec{E}(\vec{r}',t') \quad \Leftrightarrow \quad \vec{P}(\vec{\kappa},\omega) = \chi(\vec{\kappa},\omega)\,\vec{E}(\vec{\kappa},\omega)$$

This is now a convolution in space and time.
If the material is isotropic, no direction is preferred.

$$\vec{\chi} = \chi I \qquad P_x \leftrightarrow E_x \quad P_y \leftrightarrow E_y \quad P_z \leftrightarrow E_z$$

Dipoles are created by E field and in turn creates E field. But the response χ needs some models to predict.

Relations Between Lights and Materials

From the wave equations

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\vec{E}(\vec{r},t) = -\frac{4\pi}{c^2}\frac{\partial^2}{\partial t^2}\vec{P}(\vec{r},t)$$

Since $\mathcal{F}\left[\frac{\partial^2}{\partial t^2}\vec{E}\right] = -\omega^2\vec{E}$ and $\mathcal{F}[\nabla^2\vec{E}] = -\kappa^2\vec{E}$ and $\mathcal{F}[\vec{P}(\vec{r},t)] = \chi(\vec{\kappa},\omega)\vec{E}(\kappa,\omega)$

$$\left\{\frac{\omega^2}{c^2}(1+4\pi\chi(\vec{\kappa},\omega)) - \kappa^2\right\}\vec{E}(\vec{\kappa},\omega) = 0$$

from a propagating E field where material create a polarization source for a new additional generated field, we arrive at the FT. It analysis how the material reacts to different light frequencies

$$\boxed{\frac{\omega^2}{c^2}\varepsilon(\vec{\kappa},\omega) = \kappa^2} \quad \text{where } \varepsilon = 1 + 4\pi\chi(\vec{\kappa},\omega) = \tilde{n}^2 \quad \underline{\text{Dispersion}}$$

Now, assume a mono chromatic comes in the medium

$$\vec{E}(z,t) = \tfrac{1}{2}\vec{E}_0 e^{-i(\omega t-\kappa z)} + c.c$$

ω is real and κ complex.

$\text{Im}\,\kappa > 0$ decay

$\text{Im}\,\kappa < 0$ grow

Insert in wave equation gives

$$\frac{\omega^2}{c^2}\varepsilon = \kappa^2 \rightarrow \text{exactly as above and this form } e^{\pm\kappa} \text{ is a solution}$$

We learn that κ is complex when ε is complex

$$\kappa = \kappa' + i\kappa'' \qquad \varepsilon = \varepsilon' + i\varepsilon''$$

Expanding gives

$$\frac{\omega^2}{c^2}\varepsilon' = \kappa'^2 - \kappa''^2 \qquad \frac{\omega^2}{c^2}\varepsilon'' = 2\kappa'\kappa''$$

Define

$$\kappa_0 = \frac{\omega}{c} \qquad \kappa' = n\kappa_0 \qquad \alpha = 2\kappa''$$

By observing the E form, κ' plays an oscillatary role while κ'' represents the absorption

$$\vec{E}(z,t) = \frac{1}{2}\vec{E}_0 e^{-i(\omega t - n\kappa_0 z)} e^{-\alpha/2 z} + c.c$$

Solving:

$$n = \sqrt{\tfrac{1}{2}\left[\varepsilon' + \sqrt{\varepsilon'^2+\varepsilon''^2}\right]} \qquad \alpha = \frac{\omega}{cn}\varepsilon''$$

$$\varepsilon' = n^2 - \kappa^2 \qquad \varepsilon'' = 2n\kappa$$

Alternative definition

$$\tilde{n} = n + i\kappa$$

$$\tilde{n} = \sqrt{\varepsilon' + i\varepsilon''}$$

$$\kappa = \frac{\omega}{c}\tilde{n}$$

By taking intensity, we arrive right at the Beer's law

$$\text{Intensity} \sim \left|\tfrac{1}{2}\vec{E}_0 e^{-i(\omega t-\kappa z)}\right|^2$$

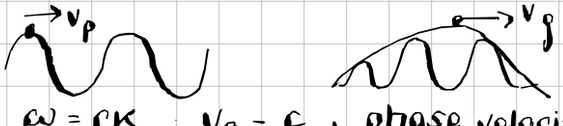
$$\sim e^{-\alpha z}$$

Consider a pulse, it has many nearby frequences $\omega_0, \omega_0 \pm \delta\omega, \ldots$ or $\kappa_0, \kappa_0 \pm \delta\kappa$. For example

$$E_1 = \cos(\kappa_1 z - \omega_1 t) \quad \text{and} \quad E_2 = \cos(\kappa_2 z - \omega_2 t)$$

$$E = 2\cos\left(\frac{\Delta\kappa z - \Delta\omega t}{2}\right)\cos(\bar{\kappa}z - \bar{\omega}t) \qquad \bar{\kappa} = \frac{\kappa_1+\kappa_2}{2}$$

$$v_p = \frac{\bar{\omega}}{\bar{\kappa}} \qquad v_g = \Delta\omega/\Delta\kappa = \frac{\partial\omega}{\partial\kappa}$$



In vacuum, $\omega = c\kappa$ $v_g = c$, phase velocity is the same as group velocity (non dispersive)

When material is dispersive, $\kappa(\omega) = \frac{\omega n(\omega)}{c} \Rightarrow v_g = \frac{c}{n + \omega\frac{dn}{d\omega}}$

The group velocity is different than phase velocity